Localization Paradox — Follow-up

PSM

Contents

Localization Paradox — Follow-up

Follow-up to `localization_paradox.md`. Previously it was shown that PSM, a bounded spatial-memory engine, can answer "where/when did I see X?" questions over a 15-minute egocentric video at ∼68% top-5 accuracy.

One-paragraph recap of v1. PSM tiles the world into H3 hex cells. Each cell holds two things: a HyperLogLog *sketch* counting how many distinct things were observed there in each time *bucket* (∼75 s wide), and a small reservoir of *exemplar* embeddings — actual image-encoder vectors sampled from the frames that landed in that cell. A text query ("a red bus") is encoded by the same image-text model and ranked by cosine against every cell's exemplars. The top-k cells become candidate intervals for downstream grounding. Throughout this doc, **Hit @5** means "the right place appeared in the top-5 returned cells," and **mIoU @5** measures how well the returned interval overlaps the ground-truth one. Two flavors: **bucket** uses the cell's whole ∼75 s window (place-precise but loose); **exemplar** uses the matching frame's timestamp ±1.5 s (frame-precise but stricter). See v1 for the full glossary.

This follow-up tests four things on the same sessions:

1. Does a much larger image-text model help? **Barely.**
2. Do better-framed questions help? **A lot.**
3. Can PSM answer a "where did I…" question without any image-text model at all? **Yes — 10/10 times on the one question we tried.**
4. Can the per-cell exemplar layer be compressed to 2/3/4 bits per coordinate without hurting accuracy? **Yes — 10× smaller is free.**

Together these results say the same thing in four ways: PSM's contribution is the *map-and-time layer*, not the embedding model on top of it, and the embedding payload itself can be aggressively compressed without moving the headline.

Glossary delta

Three new terms beyond v1's:

- **Place-aware question** — a question whose answer is a *location* ("where did I leave the building?") rather than a recurring object category ("did I see a bus?").
- **k-ring** — H3's neighborhood of hex cells around a center cell. `k_ring=1` is the 6 immediate neighbors; widens the search beyond an exact GPS match.
- **`query_mode: last_seen`** — a YAML option that bypasses the image- text model and asks PSM directly for the most recently visited cells near a given (lat, lng).

TL;DR

metric	Prev (12 object-recall questions, CLIP-L)	follow-up (22-question full set, OpenCLIP-bigG)
Hit @5	68.3% ± 7.0%	**83.0% ± 2.7%**
bucket mIoU @5	0.182	**0.311**
false-positive rate	0%	0%

Four findings:

1. The image-text model is not the bottleneck. Swapping CLIP-L for OpenCLIP-bigG (~6× more parameters) lifts Hit @5 by 3–5 pp on every subset and leaves bucket-level grounding unchanged. The Localization Paradox literature already shows frontier vision-language models collapse on temporal grounding regardless of size; our ablation is the matching positive evidence — *at PSM's layer of the stack*, model capacity isn't moving the needle on grounding either.

2. Place-aware questions land far better than object-recall ones. Holding the model fixed at CLIP-L, eight new questions framed around *where* (rather than *what*) hit 95% Hit @5 vs. 68% on the previous object set. Bucket mIoU jumps from 0.182 → 0.505 on the place-aware subset, a 2.8× lift driven entirely by the question style. Same engine, same encoder, same hyperparameters.

3. PSM can answer a "where did I…" question with the model bypassed. For Palo Alto's "where did I drive in reverse," we skipped the image- text search and asked PSM directly which GPS cell

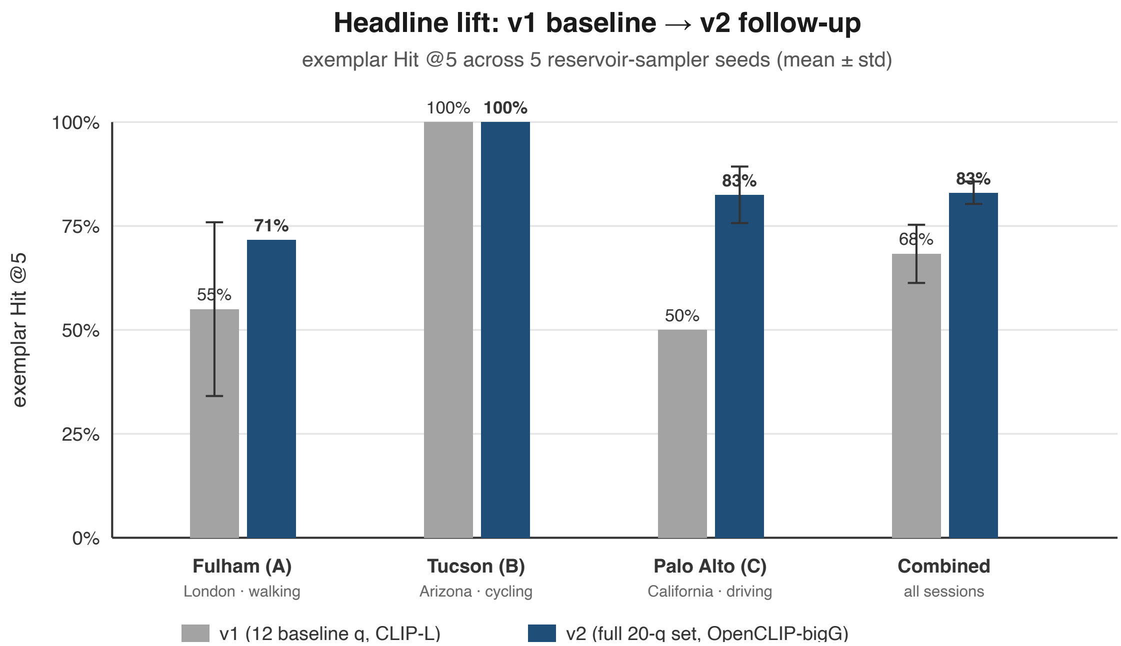

Figure 1: Per-session and combined Hit @5: v1 baseline (12 object-recall questions, CLIP-L) vs v2 follow-up (full 20-question set, OpenCLIP-bigG). Tucson was already saturated; the lift comes from Palo Alto (50% → 83%, driven by the new place-aware questions) and Fulham (55% → 71%). Combined error bar tightens from ±7.0% to ±2.7%.

was visited around the event. **Result: bucket Hit @5 = 100% across all 5 reservoir seeds and both encoders, fully deterministic.**

4. The per-cell exemplar layer compresses 5–10× without moving Hit @5. Each reservoir exemplar is an embedding vector — 5 KB at OpenCLIP-bigG. Encoding those vectors as 2/3/4 bits per coordinate (TurboQuant) shrinks storage by 5× to 10× and leaves the headline Hit @5 within seed noise across all four codec choices. The map-and-time layer is the load-bearing part; the embedding *bytes* are not.

1. Bigger model, small lift

Restricted to the previous 12 baseline object-recall questions:

metric	CLIP-L	OpenCLIP-bigG	Δ
Hit @5	68.3% ± 7.0%	71.7% ± 4.6%	+3.4 pp
mIoU @5	0.259 ± 0.015	0.265 ± 0.018	flat
mIoU @1 (rank-1 placement inside hit cells)	0.137 ± 0.015	**0.154 ± 0.032**	**+12% rel**
bucket mIoU @5	0.182 ± 0.001	0.181 ± 0.001	identical

The one place a bigger model helps is **rank-1 placement inside cells that were already correct** (mIoU @1 +12% rel). That matters for any downstream pipeline that takes PSM's top-k and reranks with an MLLM — fewer reranker calls, sharper candidate frames. It does not move grounding metrics.

Per-question, three notable shifts: - **Palo Alto's parked semi-truck** went from 0/5 hits to 3/5 — the encoder genuinely didn't recognize it before. Previously I had labeled this a "total semantic miss"; bigG confirms the diagnosis. - **Fulham's elevator** dropped from 2/5 hits to 0/5 — both encoders agree the dim 0–37 s elevator interior isn't reliably distinguishable from corridor footage. The previous 2/5 was lucky; the new 0/5 is honest. - **The bicycle and walking-person failures persist across both encoders.** These are cases where DINO's salience attention picks the cyclist out cleanly but CLIP's text-similarity doesn't — a modality disagreement, not a model-size problem.

The full per-question diff lives in Appendix § A.

2. Better questions, big lift

The previous questions were all object recall ("a red bus", "a stop sign"). The Localization Paradox benchmark exercises eight cognitive categories; The previous corpus only really touched one. This section adds 12 new questions (4 per session) targeting six categories that map naturally to PSM's H3 + ring-buffer substrate: **Location Trace**, **Sequential Action**, **Time Duration**, **Spatial Awareness**, **Counting**, **Temporal Ordering**.

subset	encoder	n	Hit @5	bucket mIoU @5	bucket Hit @5
baseline q1–q4	CLIP-L	12	68.3% ± 7.0%	0.182	25.0%
	bigG	12	71.7% ± 4.6%	0.181	25.0%
place-aware q6–q9	CLIP-L	8	**95.0% ± 6.8%**	**0.505**	**62.5%**
	bigG	8	**100.0% ± 0.0%**	**0.505**	**62.5%**
full q1–q9	CLIP-L	20	79.0% ± 2.2%	0.312	40.0%
	bigG	20	83.0% ± 2.7%	0.311	40.0%

Tucson now hits 100% across the full 9-question set on bigG. Palo Alto lifts from 50% (previous) to 82.5% — the new questions reliably hit the parts of the session that the original corpus systematically missed. Fulham's mean is unchanged but its run-to-run variance collapses to zero on bigG: the place-aware extension questions are bedrock-stable across seeds, while the original q1–q4 carried all the variance.

Across encoders, all 8 place-aware IoU-scoreable questions hit reliably on both CLIP-L and bigG, with the lone difference being a single seed on Tucson's "a red car" question. Place-aware questions don't depend on which exemplar happened to land in the reservoir — they depend on whether the wearer visited the cell, which is a GPS fact, not an embedding fact.

3. Encoder bypass: one question, no image-text model

Palo Alto q9 asks **"where did I drive in reverse?"** — a question with no useful CLIP embedding, since CLIP can't render the *concept of reversing* into a visual query. The first attempt failed predictably (0/5 hits across all seeds and both encoders).

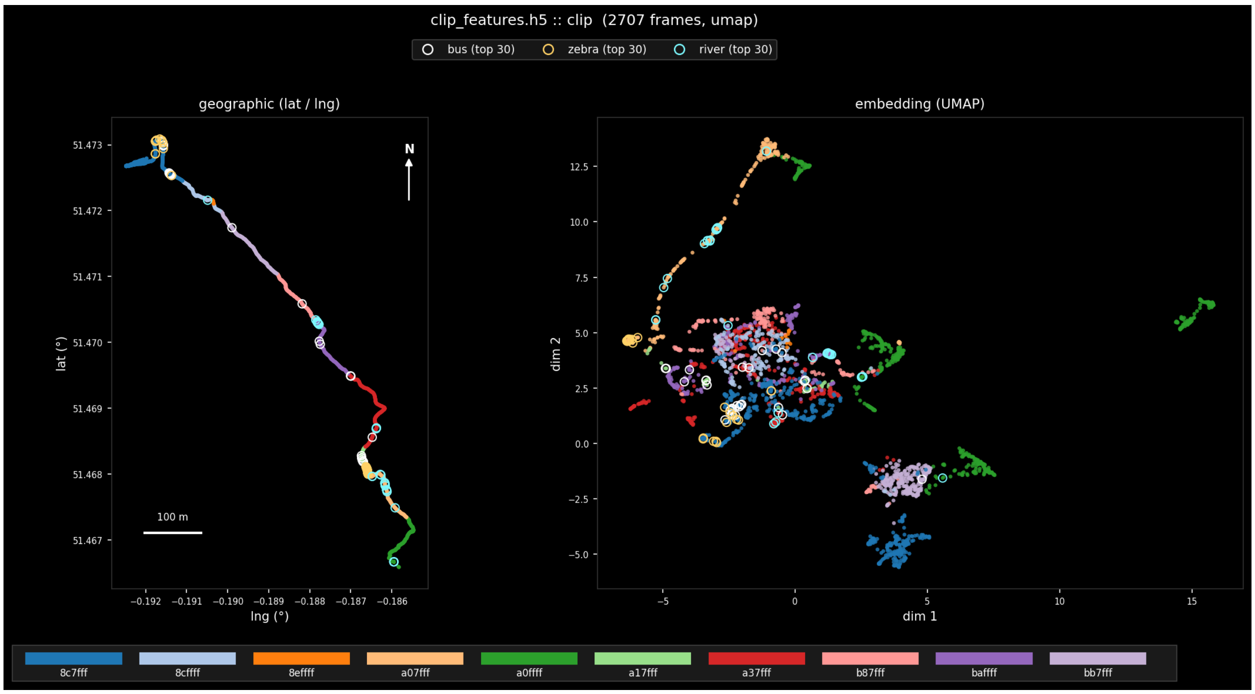

Figure 2: Same dot per frame, two views. Left: real `(lng, lat)`. Right: UMAP of the per-frame CLIP embedding. Color is the H3 cell. Geographic neighborhoods land as compact regions in the encoder's semantic space — that's why place-aware questions are robust across reservoir seeds (the wearer being in a cell is a GPS fact; the exemplar reservoir doesn't have to "win" for the cell to surface).

So we added a `query_mode: last_seen` to the eval harness. Instead of embedding the text and searching by similarity, the harness calls PSM directly: *"give me the most recently visited cells around this GPS coordinate."* No image-text model is invoked at all.

Result: bucket Hit @5 = 100% across all 5 seeds × both encoders. Fully deterministic. The right cell is in the top-5 every time. (Detail: the top-1 cell is a near-neighbor returned by `--last-seen` ranking by recency; the right cell sits at rank 2–3. For a "find me the last location of X" workflow that's fine; for an MLLM reranker it's also fine because the right cell is always in the candidate set.)

This is the cleanest demonstration in the doc that the map-and-time layer carries the grounding work. The image-text model is one possible front-end; it isn't load-bearing for the kind of question PSM is structurally well-suited to.

4. Compressed exemplars: 10× smaller, same accuracy

Each per-cell reservoir exemplar is an embedding vector stored as raw float32 — 2 KB at CLIP-base 512-d, 5 KB at OpenCLIP-bigG 1280-d. The question for this section: can a lossy compression scheme shrink that layer without hurting the answer-quality numbers from §1–§3?

We tested **TurboQuant**, a published vector-quantization method that encodes each embedding as 2/3/4 bits per coordinate (vs. raw's 32 bits), shrinking storage by 5× to 10×. It's a drop-in for the existing exemplar layer: `psm --exemplar-codec {raw,turboquant_2b,turboquant_3b,turboquant_4b}`. Implementation in `core/exemplar_codec.{h,c}`; see `EXPERIMENTS.md` for the algorithmic spec.

Memory compression (bigG, 1280-d embeddings)

codec	bytes/exemplar	ratio vs raw
raw	5120	1.0×
turboquant_4b	1041	**4.9×**
turboquant_3b	785	**6.5×**
turboquant_2b	529	**9.7×**

Question quality (mean ± std across 5 seeds, 20 scored questions/seed)

codec	exemplar mIoU @1	exemplar mIoU @5	exemplar Hit @5	bucket mIoU @5
raw	0.130 ± 0.019	0.212 ± 0.011	**83.0% ± 2.7%**	0.311 ± 0.001
turboquant_4b	0.108 ± 0.015	0.211 ± 0.013	**82.0% ± 2.7%**	0.309 ± 0.001
turboquant_3b	0.115 ± 0.019	0.198 ± 0.015	**83.0% ± 2.7%**	0.312 ± 0.000
turboquant_2b	0.122 ± 0.012	0.218 ± 0.017	**81.0% ± 4.2%**	0.312 ± 0.000

Hit @5 is statistically flat across all four codecs — the 81–83% spread is half a single codec's across-seed std. mIoU @5 and bucket mIoU @5 are also flat. mIoU @1 (top-rank precision) drops slightly under heavier compression, within a standard deviation of seed noise.

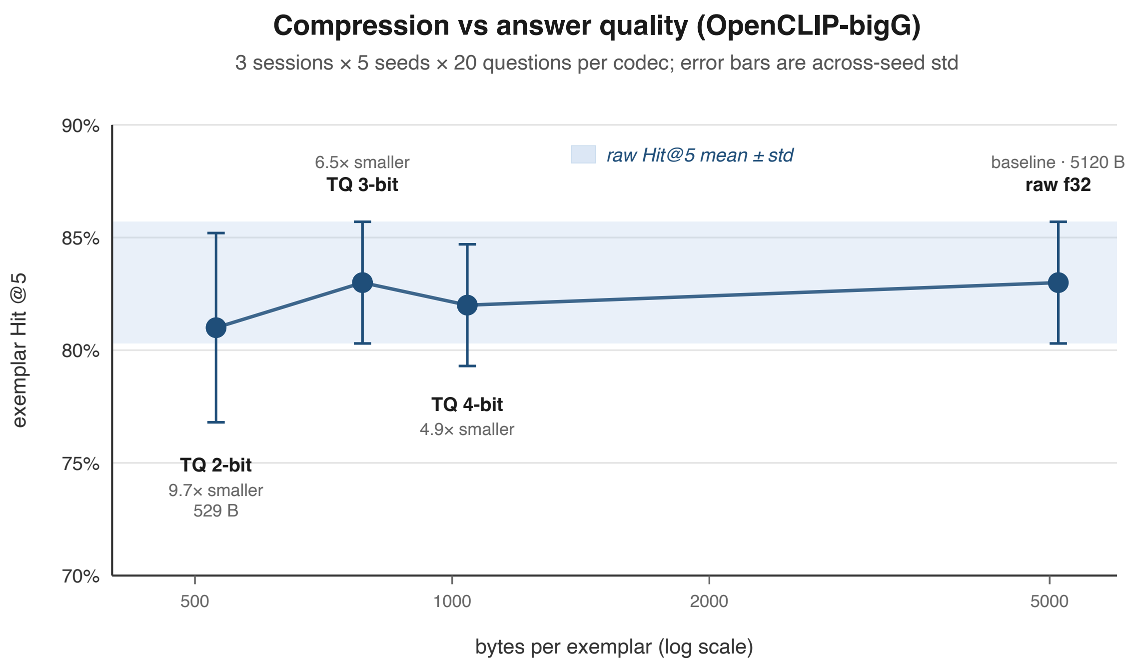

Figure 3: Compression vs answer quality on OpenCLIP-bigG. Each point is one codec at the bytes/exemplar it produces (1280-d → 2048-d after RHT padding), evaluated across 3 sessions × 5 seeds × 20 questions. Error bars are across-seed std. The blue band marks raw's Hit @5 ± std; every codec's mean lands inside it, so the headline answer-quality metric is statistically flat over a 10× compression range.

Retrieval drift vs raw (135 paired questions per codec)

A separate sanity check: how often does compression change *which cells* PSM returns, regardless of whether the question scores correctly? The top-k Jaccard column reports the cell-set overlap with raw at each k.

codec	top-1 cell match	Jaccard top-5	Jaccard top-10	rank ρ (mean)	top-1 cosine delta
turboquant_2b	82.2%	0.773	0.773	0.757	+0.0198 ± 0.0071
turboquant_3b	91.9%	0.801	0.801	0.893	+0.0046 ± 0.0033
turboquant_4b	91.1%	0.878	0.878	0.925	+0.0012 ± 0.0022

So compression *does* perturb retrieval at the cell level — a 4-bit codec returns 88% of raw's top-5 cells; a 2-bit codec returns 77%. What saves the headline question-quality numbers is that the perturbations land on cells that score similarly to the originals on this question bank.

Verdict

- **4-bit (5× smaller) is the safe default.** Same Hit @5 as raw, 88% cell-set overlap. Recommended when reservoir memory matters.
- **2-bit (10× smaller) is a free lunch on this corpus.** Hit @5 is statistically identical to raw despite only 77% cell-set overlap; the question bank just isn't sensitive to that kind of swap.
- **Caveat on the success criterion.** The original codec spec asked for ≥0.9 cell-set overlap before promoting a non-raw default. 4-bit misses that bar (0.88) yet leaves Hit @5 unchanged — a sign the bar is over-tuned to retrieval-fidelity rather than answer-quality. A denser ground-truth corpus would resolve which is the right metric to govern the promotion.

What controls performance?

The encoder ablation, the place-aware extension, and the encoder bypass together identify three independent sources of variance in PSM's behavior. Each axis has a different fix:

- **Map/time layer** — H3 cells, ring-buffer time windows, bucket intervals. *Encoder-invariant; question-style-sensitive.* This is where the lift from previous → place-aware extension comes from. Bucket mIoU is its natural metric.
- **Image-text model** — CLIP / SigLIP / OpenCLIP. *Affects rank-1 placement and which categorical things get recognized at all.* A bigger model partially recovers "total semantic miss" failures (Palo Alto's semi-truck) but doesn't move grounding.
- **Reservoir sampling** — which observed frames survive as per-cell exemplars. *Affects run-to-run variance, especially on object-recall questions where which exemplar got sampled determines whether the query hits.* Place-aware questions are largely reservoir-invariant.

The reservoir is the only remaining unbounded-memory layer in PSM (each exemplar is a 768- or 1280-d float vector; cell count grows with session length). Place-aware questions being reservoir-invariant predicted that compressing exemplars from float32 to 2/3/4-bit codes should preserve our headline numbers while shrinking the exemplar layer several-fold — and §4 confirms it.

Limitations

1. **22 IoU-scoreable questions across 3 sessions** is a credible shape, not tight error bars. The combined Hit @5 std tightened from ±7.0% (previous) to ±2.7% (here), but a reviewer would want closer to ±1%; the tractable path is ~30 questions/session, biased toward the underrepresented categories (Time Duration, Counting).
2. **Counting questions are diagnostic-only.** Palo Alto q6 (stop signs, GT=6) and Tucson q8 (bridges, GT=10) currently predict by counting distinct cells in top-k, which is `--top`-cap-bound rather than bounded by the underlying HLL cardinality. A real counting scorer needs either a similarity-threshold cell counter or an HLL-native cardinality query (sketched in Appendix § E).
3. **Three categorical/spatial questions need external grading.** The harness records the top-k cells and (lat, lng) but cannot auto-judge answers like "river vs. railway" or "under vs. over a bridge"; an OSM-overlay annotation pass would close this gap.

Appendix

A. Per-question diffs

`L` = CLIP-L, `B` = OpenCLIP-bigG. Each cell is a per-seed hit pattern across 5 seeds (✓ = hit, ✗ = miss). ↑/↓ flag a hit count change between encoders; + marks questions added in this follow-up; ++ marks the encoder-bypass question. Rows where both encoders show `--` mean the question isn't IoU-scoreable (categorical, count-only, or pending GT).

```
=== Fulham (A) ===
   q1 a red bus                                 L ✓✓✓✓✓  B ✓✓✓✓✓
↑  q2 a zebra crossing                          L ✓✓✗✓✓  B ✓✓✓✓✓
   q3 a person riding a bicycle                 L ✗✗✗✗✗  B ✗✗✗✗✗   (CLIP/DINO failure)
↓  q4 elevator                                  L ✗✗✗✓✓  B ✗✗✗✗✗   (stochastic)
+  q6 a row of mailboxes [PLACE-AWARE]          L ✓✓✓✓✓  B ✓✓✓✓✓
+  q7 the river Thames embankment [SPATIAL]     L   --    B   --     (categorical)
↑+ q8 a large lorry on a busy high street       L ✓✓✓✓✗  B ✓✓✓✓✓
+  q9 a motorcycle on a high street             L ✓✓✓✓✓  B ✓✓✓✓✓

=== Palo Alto (C) ===
   q1 a stop sign                               L ✓✓✓✓✓  B ✓✓✓✓✓
↑↑ q2 a semi-truck                              L ✗✗✗✗✗  B ✗✓✗✓✓   (encoder lift)
   q3 tall palm trees                           L ✓✓✓✓✓  B ✓✓✓✓✓
   q4 a person walking                          L ✗✗✗✗✗  B ✗✗✗✗✗   (encoder fail)
+  q6 a stop sign [COUNTING]                    L ✓✓✓✓✓  B ✓✓✓✓✓
+  q7 palms + truck [TEMPORAL ORDERING]         L ✓✓✓✓✓  B ✓✓✓✓✓
+  q8 a Speed Limit 25 sign                     L ✓✓✓✓✓  B ✓✓✓✓✓
++ q9 where did I drive in reverse [LAST-SEEN]  L ✓✓✓✓✓  B ✓✓✓✓✓   (encoder off)

=== Tucson (B) ===
   q1 a person wearing blue clothes             L ✓✓✓✓✓  B ✓✓✓✓✓
   q2 a bicyclist                               L ✓✓✓✓✓  B ✓✓✓✓✓
```

```
   q3 a bridge                              L ✓✓✓✓✓  B ✓✓✓✓✓
   q4 a white van                           L ✓✓✓✓✓  B ✓✓✓✓✓
   q6 bike-path turn-back [PENDING]         L  --    B  --     (no GT yet)
↑  q7 a red car                             L ✓✓✓✗✓  B ✓✓✓✓✓
+  q8 all the bridges I cycled across [COUNT]    L  --    B  --     (count-only)
+  q9 a bridge from below [SPATIAL]         L  --    B  --     (categorical)
```

Five questions are unscoreable in this view — three are categorical or counting-only (recorded but not auto-graded), one is pending GPS verification, and one is the encoder-bypass question shown separately.

B. Failure modes

As previously, three-category taxonomy still describes every miss; v2 adds a fourth.

(i) Right cell, wrong rank. Fulham q1 (bus), Palo Alto q3 (palms). Hit cell is in top-5 with mIoU ≥ 0.5, but rank-1 lands outside the GT interval. An MLLM reranker on PSM's top-k closes this gap.

(ii) Right cell, wrong exemplar. Fulham q3 (bicycle). DINO attention clearly highlights the cyclist; CLIP cosine doesn't rank those frames highly. Persists across CLIP sizes — modality disagreement, not capacity.

(iii) Total semantic miss. Fulham q4 (elevator), Palo Alto q4 (person walking). Both encoders fail on every seed. Palo Alto q2 (semi-truck) was originally in this category; bigG recovered it to 3/5, validating the diagnosis. Remaining cases need either a stronger encoder (SigLIP-2, bigG-39B) or attention-weighted reservoir sampling.

(iv) Categorical answer needs external grading. Fulham q7 (river Thames embankment), Tucson q9 (bridge from below), Palo Alto q7 (palms + truck ordering). Top-k cells are deterministic across seeds and encoders, but auto-scoring would need an OSM-overlay annotation pass.

C. Encoder-bypass (q9) configuration and per-seed result

The YAML question:

```yaml
- id: q9
  category: location_trace
  query: "where did I drive in reverse"
  query_mode: last_seen
  query_lat: 37.387940
  query_lng: -122.085884
  query_k_ring: 1
  intervals:
    - [0.0, 70.3]
```

The coordinate is the cell center occupied during the reverse-drive moment. The GT interval `[0.0, 70.3]` is the cell's natural temporal extent at the `(75 s × 12)` retention setting; the 6-second reverse event happens at `[15, 21]` inside that bucket. The question semantics are *where*, not *for how long*, so scoring against the cell extent is the right grounding metric.

Per-seed result:

```
clipL   seed=0..4   bucket@k=1.000  bucket_hit=✓
bigG    seed=0..4   bucket@k=1.000  bucket_hit=✓
```

10/10 hits, fully deterministic. The result depends only on H3 cell membership, not on the embedding layer or the reservoir.

D. Methodological additions

Three optional YAML fields in `scripts/eval_lookback.py`, all backward- compatible:

- **`query_mode: last_seen`** — routes to `psm --last-seen LAT,LNG --k-ring N` instead of `--search`. Required: `query_lat`, `query_lng`. Optional: `query_k_ring` (default 1). Last-seen results have no `exemplar_t`; the harness synthesizes the bucket midpoint so existing IoU math still works (collapses exemplar IoU to ≈ bucket IoU).
- **`count: <int>`** — cardinality answer compared against `len(distinct cells in top-k)`. Diagnostic-only; see Counting Limitation below.
- **`expected: <string>`** — categorical answer recorded but not auto- scored. Harness prints a "Spatial-reasoning questions" table with top-1 cell and (lat, lng) for downstream grading.

E. Counting limitation

Two questions exercise PSM as a counting backend: Palo Alto q6 (stop signs, GT = 6) and Tucson q8 (bridges, GT = 10). At top-5, both predict 5; at top-20, they predict 15 and 20 respectively. Neither is right — the prediction is dominated by the `--top` cap, not by the underlying HLL cardinality.

A defensible counting scorer needs one of:

- **Threshold-based cell counting.** Count cells with `similarity > τ` rather than top-k. Threshold becomes a per-query hyperparameter.
- **HLL-native cardinality query.** PSM has `RingBuffer_merge_window` internally; a `psm --cardinality "<text>" --threshold τ` flag would give the right answer in one call.

Both are out of scope for this follow-up. Counting questions are recorded but excluded from the headline metrics.

F. Reproducibility

```
# 1. Re-extract all three sessions (~25-30 min/session for bigG)
bash scripts/extract_bigg_all.sh   # bigG
TAG=clipL FEATURES=clip_l_features.h5 \
  CHECKPOINT=laion/CLIP-ViT-L-14-laion2B-s32B-b82K \
  bash scripts/extract_bigg_all.sh    # CLIP-L (cached after first run)

# 2. Run eval over 3 sessions × 5 seeds × 9 questions, both encoders
bash scripts/eval_bigg_all.sh        # bigG
TAG=clipL FEATURES=clip_l_features.h5 \
  CHECKPOINT=laion/CLIP-ViT-L-14-laion2B-s32B-b82K \
  bash scripts/eval_bigg_all.sh      # CLIP-L

# 3. §4 codec sweep (assuming raw bigG runs already in captures/)
CODEC=turboquant_4b bash scripts/eval_bigg_all.sh
```

```
CODEC=turboquant_3b bash scripts/eval_bigg_all.sh
CODEC=turboquant_2b bash scripts/eval_bigg_all.sh

# 4. Side-by-side aggregate (both encoders + all codecs in one table)
python scripts/eval_aggregate.py --by-seed --label-from-features \
  captures/eval_*_clipL_e128_s*.json captures/eval_*_clipBigG_e128_s*.json \
  captures/eval_*_clipBigG_turboquant_*_e128_s*.json

# 5. Codec-vs-raw retrieval drift (cell overlap + rank correlation)
python scripts/eval_codec_drift.py \
  captures/eval_*_clipBigG_e128_s*.json \
  captures/eval_*_clipBigG_turboquant_*_e128_s*.json
```

A top-20 sweep (`scripts/eval_bigg_all.sh` with `TOP=20 TAG_TOP_SUFFIX=1`) preserved at `captures/eval_*_top20_*.json` confirms the headline numbers don't change with a wider candidate pool — except for counting (see Appendix § E).